

GOKTUG MERT OZDOGAN

Software Engineer | AI/ML and Backend Systems | Building AI-Driven Products

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PROFESSIONAL SUMMARY

Software Engineer specializing in AI-integrated full-stack applications, backend systems, and cloud deployment. Independently built and shipped Nummoria, a cross-platform AI-powered personal finance SaaS (Node.js, React, React Native, MongoDB, LLM APIs). Currently contributing to an agentic AI system for scientific paper reproducibility through inzva AI Projects #10. Previously delivered production features supporting 10,000+ employees at a major Turkish bank using C#.NET Core and Angular. Strong foundation in Python, machine learning, REST APIs, system design, Docker, and AWS.

PROFESSIONAL EXPERIENCE

Halkbank — One of Turkey's Largest State-Owned Banks

04/2025 – 07/2025

Full-Stack Software Engineering Intern

Istanbul, Turkey

- Implemented a Personnel Absence System serving **10,000+ employees**, building attendance-tracking logic in C#.NET Core across **75+ relational tables** covering holidays, medical leave, and approval workflows.
- Developed full-stack reporting workflows in Angular and ASP.NET Core, supporting real-time data processing across **800+ bank branches** throughout Turkey.
- Owned all **4 stages** of production feature delivery — implementation, testing, code review, and deployment — within an enterprise Agile/Scrum environment.

ComPro — IBM Platinum Partner

08/2023 – 10/2023

Software Engineering Intern

Istanbul, Turkey

- Containerized legacy backend environments with **Docker, CI/CD pipelines, and Linux**, supporting 3 delivery tracks — cloud-migration proofs of concept, internal development, and client demonstrations.
- Evaluated **AWS and IBM Cloud** for containerized workloads, comparing compute (EC2 vs. Virtual Servers), storage (S3 vs. Object Storage), orchestration, cost, and performance to support migration strategy decisions.

PROJECTS

ReproBot — Agentic AI for Scientific Reproducibility

06/2026 – Present

AI Research Contributor, inzva AI Projects #10

Istanbul, Turkey

- Co-designing a **4-agent AI pipeline** coordinated by a central Orchestrator to parse ML papers, generate experimental code, execute workflows in isolated Docker environments, and verify reproduced results against published claims.
- Conducting literature review, agent-architecture design, codebase research, and implementation planning for the **4 agent components (Reader, Coder, Runner, Critic)** using Python, vision-language models, PyTorch/Hugging Face, and structured shared memory.
- Defining an evaluation framework for a planned **20-paper image-classification benchmark**, measuring execution success, replication rate, reproduced-versus-reported metric gaps, and the impact of the iterative Critic–Coder refinement loop.

Nummoria — AI-Powered Personal Finance SaaS

10/2025 – Present

Full-Stack Engineer & Product Owner

Istanbul, Turkey

nummoria.com • iOS App: TestFlight Release

- Solely architected, developed, and deployed a personal finance SaaS across 2 platforms (Web and iOS) using Node.js, React, React Native, and MongoDB, owning the **complete product lifecycle** from system design to TestFlight release.
- Engineered modular REST APIs and optimized MongoDB schemas for transactions, categories, accounts, investment portfolios, and multi-currency financial workflows; validated support for **1,000+ concurrent users** through load testing, connection pooling, and database query optimization on a horizontally scalable backend.
- Integrated **OpenAI and Gemini APIs** to analyze user spending patterns and generate contextual personal finance insights and recommendations.
- Reduced manual transaction entry by **approximately 70%** by building an OCR-based ingestion pipeline for automated receipt and bank statement parsing.

- Secured authentication and session management with **Google Sign-In, Apple Sign-In, JWT, and OAuth 2.0**, aligned with App Store security requirements.

Eyehub — TÜBİTAK Government Research Project 122E085

11/2023 – 06/2024

Backend Engineer

Istanbul, Turkey

- Built secure **Node.js REST APIs** for a TÜBİTAK-funded dyslexia detection application, supporting mobile synchronization, authentication, real-time data submission, and academic research workflows.
- Designed and deployed AWS infrastructure on **EC2 and Lambda** to support reliable, concurrent data ingestion.
- Modeled medical research data in **MongoDB** with validation and data integrity controls for academic analysis and compliance requirements.
- Collaborated directly with project lead **Prof. Günet Eroğlu** and the university research team to translate academic and application requirements into **production-ready backend workflows and deliver project milestones**.

EDUCATION

Bahcesehir University

10/2021 – 08/2025

B.Sc. Computer Engineering

Istanbul, Turkey

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript (ES6+), C#, SQL

AI & Machine Learning: PyTorch, scikit-learn, XGBoost, Hugging Face Transformers, NumPy, Pandas, model calibration, model evaluation & error analysis

LLM & Agentic Systems: RAG pipelines, vector search (pgvector, FAISS), LangChain, multi-agent orchestration, tool-using agents, structured outputs/function calling, prompt evaluation, OpenAI & Gemini APIs

Backend & Full-Stack: Node.js/Express, FastAPI, ASP.NET Core, React, React Native, Angular, REST API design, microservices, OAuth 2.0, JWT

Data & Infrastructure: PostgreSQL, MongoDB, SQL Server, Docker, AWS (EC2, Lambda, S3), RabbitMQ, CI/CD, Linux

LANGUAGES

Turkish (Native) • English (Full Professional Proficiency) • German (B2 — Certification Prep)